

Logic PhD Exam, August 2022

Solve 5 of the following problems; at least one from each section.

A. Set Theory.

1. Let X be an uncountable set and let $F : X \rightarrow X$ be a function. Show that there are distinct elements $x, y \in X$ such that $x \neq F(y)$ and $y \neq F(x)$.
2. Let P be a forcing. Show that there is an uncountable cardinal κ such that $P \Vdash \check{\kappa}$ is a cardinal.
3. Provide an example of a subset of a Polish space which is analytic and not Borel. Provide a proof of these two properties of your set.

B. Computability.

1. Give an example of a subset of $\mathbb{N}$ that is Π_1^0 and not Σ_1^0 (and prove that your set has these two properties).
2. Prove that there exist incomparable Turing degrees.
3. Prove that there exist disjoint computably enumerable sets $A, B \subseteq \mathbb{N}$ such that there is no computable set C which contains A and is disjoint from B .

C. Model Theory.

1. Let $\mathcal{L}$ be a language comprised of a single binary relation symbol. Give an example of a countably infinite $\mathcal{L}$ -structure $\mathcal{M}$ such that. Make sure to prove both statements (i) and (ii).
 - (i) $\text{Th}(\mathcal{M})$ is $\aleph_0$ -categorical, and
 - (ii) $\text{Th}(\mathcal{M})$ does not have quantifier elimination.
2. Let $\mathcal{L}$ be a language comprised of two unary relation symbols G and B , and a binary relation symbol R . Let $\mathcal{K}$ be the class of all finite graphs with a bipartition, i.e., $\mathcal{K}$ consists of all finite $\mathcal{L}$ -structures $\mathcal{M} = (M, G^{\mathcal{M}}, B^{\mathcal{M}}, R^{\mathcal{M}})$ such that. Prove or disprove: $\mathcal{K}$ is a Fraïssé class.
 - (i) $R^{\mathcal{M}}$ is a symmetric and irreflexive relation on M ,
 - (ii) $G^{\mathcal{M}}$ and $B^{\mathcal{M}}$ form a partition of M ,
 - (iii) if $(x, y) \in R^{\mathcal{M}}$ then exactly one of x, y belongs to $G^{\mathcal{M}}$ and the other belongs to $B^{\mathcal{M}}$.
3. Let $\mathcal{L} = \{R\}$ be the language of graphs (i.e., R is a binary relation symbol). Let $\mathcal{K}_0$ be the class of all connected graphs, and let $\mathcal{K}_1$ be the class of all disconnected graphs. Prove or disprove each of the following:
 - (i) $\mathcal{K}_0$ is an elementary class;
 - (ii) $\mathcal{K}_1$ is an elementary class.